

Artificial Intelligence

For HEDSPI Project

Lecturer 4 - Search

Lecturers :

Dr. Le Thanh Huong

Dr. Tran Duc Khanh

Dr. Hai V. Pham

School of ICT, HUST

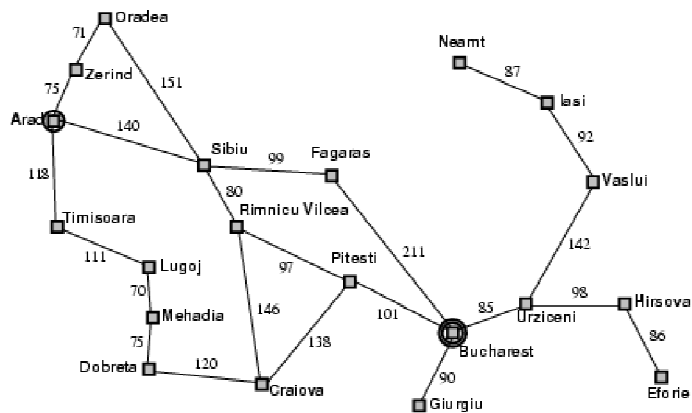
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Outline

- Graph search
- Best-first search
- A* search

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Graph search

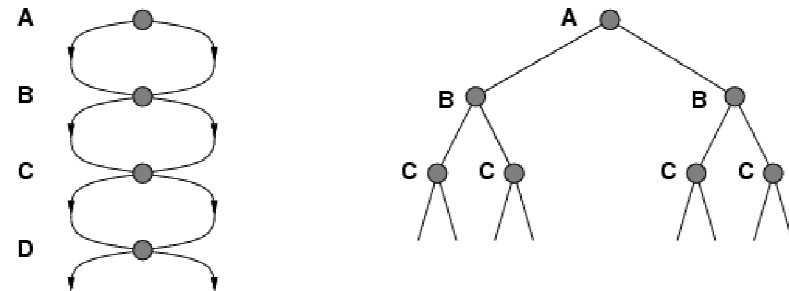


Get from Arad to Bucharest as quickly as possible

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Graph search

- Failure to detect repeated states can turn a linear problem into an exponential one!



- Very simple fix: never expand a node twice

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Graph search

```

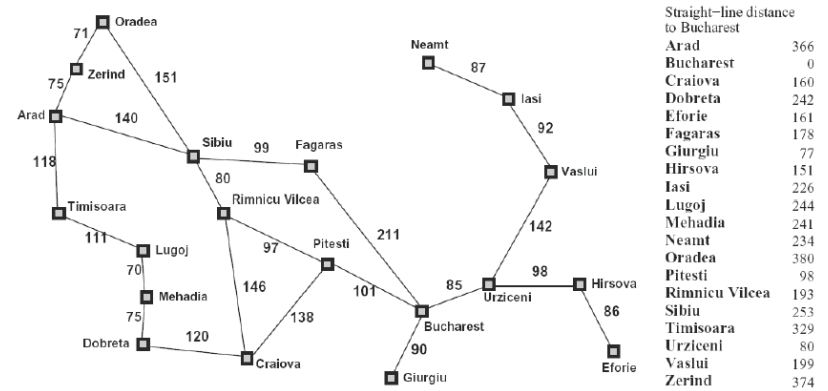
function Graph-Search(problem, fringe) returns a solution, or failure
  fringe ← Insert(Make-Node(Initial-State(problem)), fringe);
  closed ← an empty set
  while (fringe not empty)
    node ← RemoveFirst(fringe);
    if (Goal-Test(problem, State(node))) then return Solution(node);
    if (State(node) is not in closed then
      add State(node) to closed
      fringe ← InsertAll(Expand(node, problem), fringe);
    end if
  end
  return failure;

```

- Never expand a node twice!

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Straight Line Distances

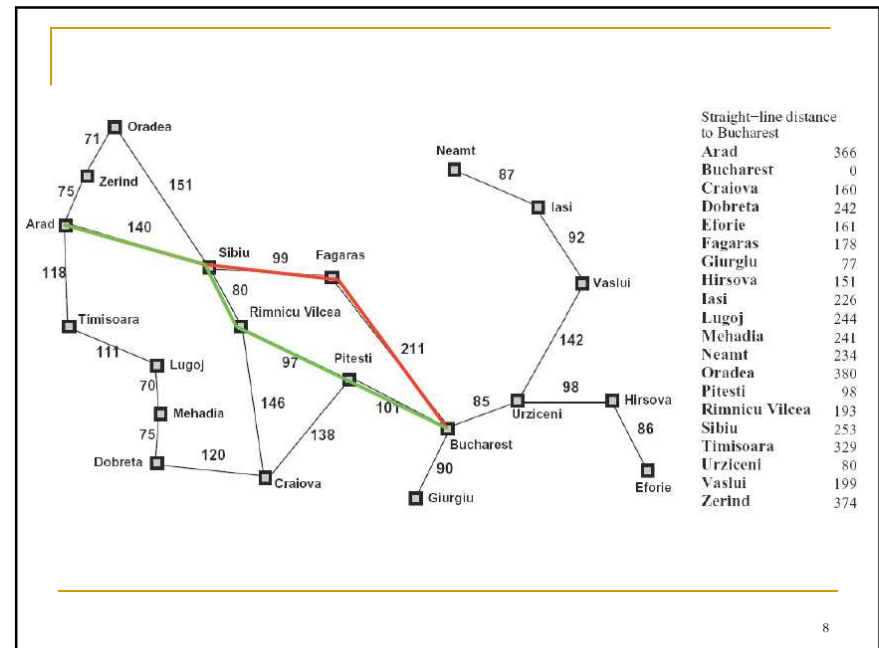


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Best-first search

- Idea: use an **evaluation function** $f(n)$ for each node
 - estimate of "desirability"
 - Expand most desirable unexpanded node
- Order the nodes in fringe in decreasing order of desirability
- Special cases:
 - greedy best-first search
 - A* search

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Greedy Best-First Search

- Evaluation function $f(n) = h(n)$ (**h**euristic)
= estimate of cost from n to *goal*
- e.g., $h_{SLD}(n)$ = straight-line distance from n to Bucharest
- Greedy best-first search expands the node that **appears** to be closest to goal

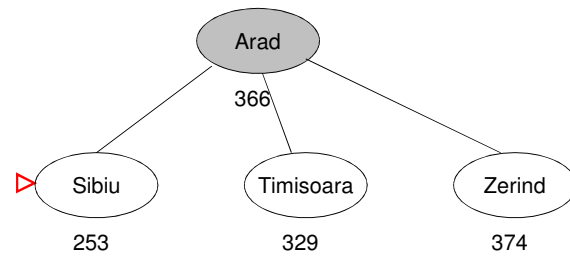
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Greedy best-first search example



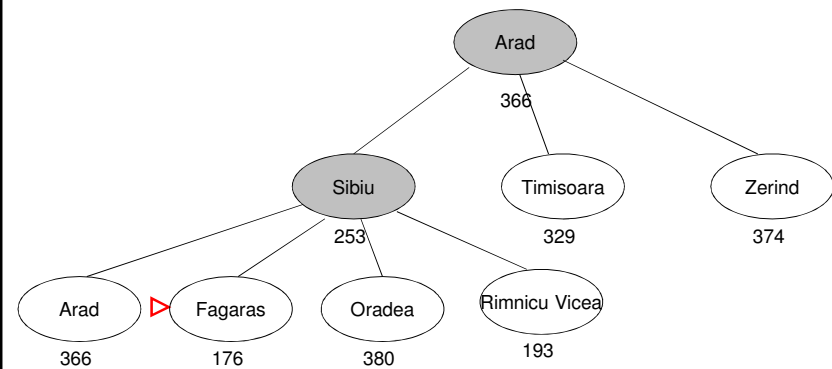
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Greedy best-first search example



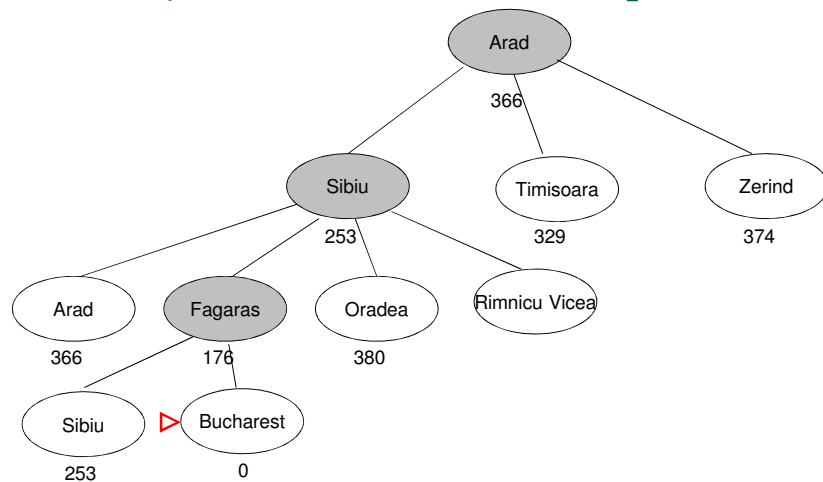
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Greedy best-first search example



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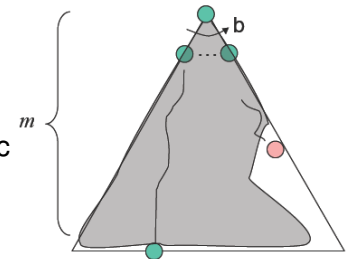
Greedy best-first search example



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Greedy Best-First Search

- **Complete?** No – can get stuck in loops, e.g., Iasi → Neamt → Iasi → Neamt → ...
- **Time?** $O(b^m)$, but a good heuristic can give dramatic improvement
- **Space?** $O(b^m)$ -- keeps all nodes in memory
- **Optimal?** No
- What do we need to do to make it complete?
⇒ A* search
- Can we make it optimal? → No



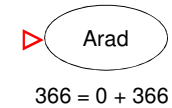
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A* search

- Idea: Expand unexpanded node with lowest evaluation value
- Evaluation function $f(n) = g(n) + h(n)$
- $g(n)$ = cost so far to reach n
- $h(n)$ = estimated cost from n to goal
- $f(n)$ = estimated total cost of path through n to goal
- Nodes are ordered according to $f(n)$.

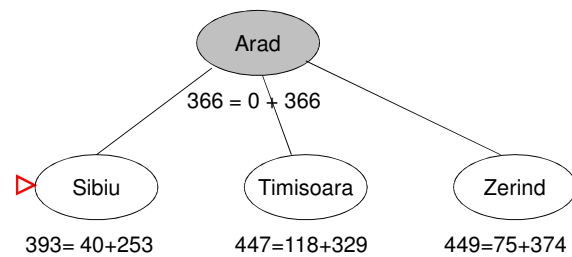
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A* search example



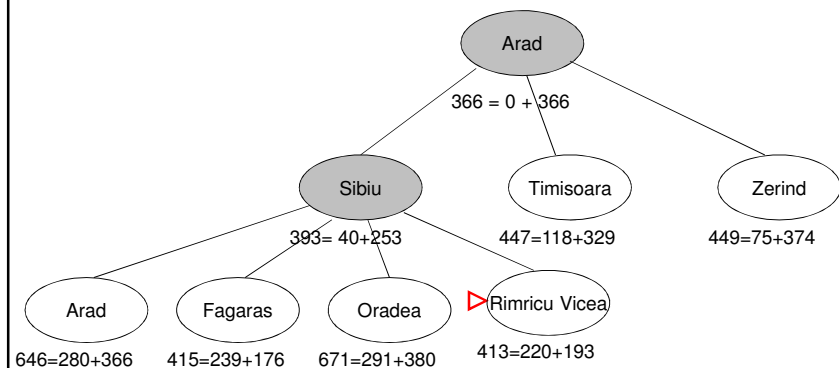
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A* search example



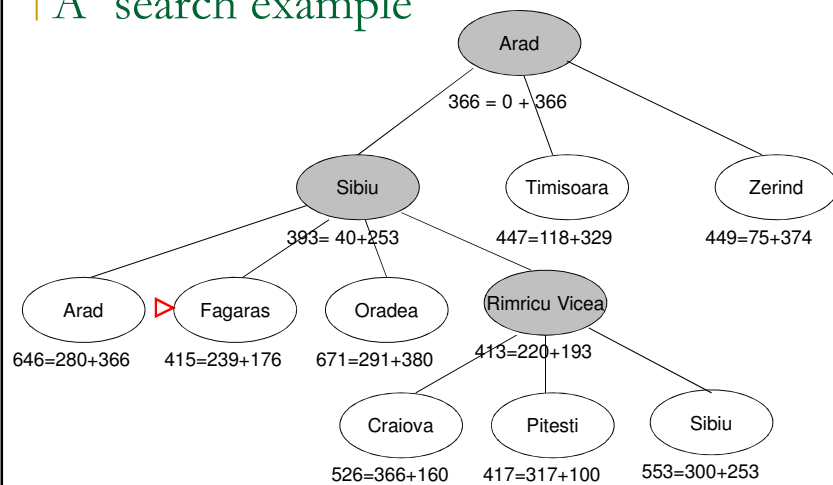
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A* search example



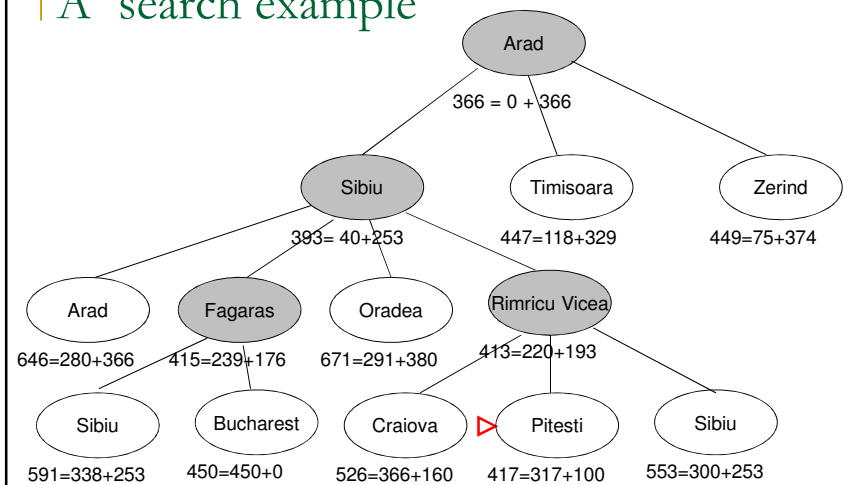
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A* search example



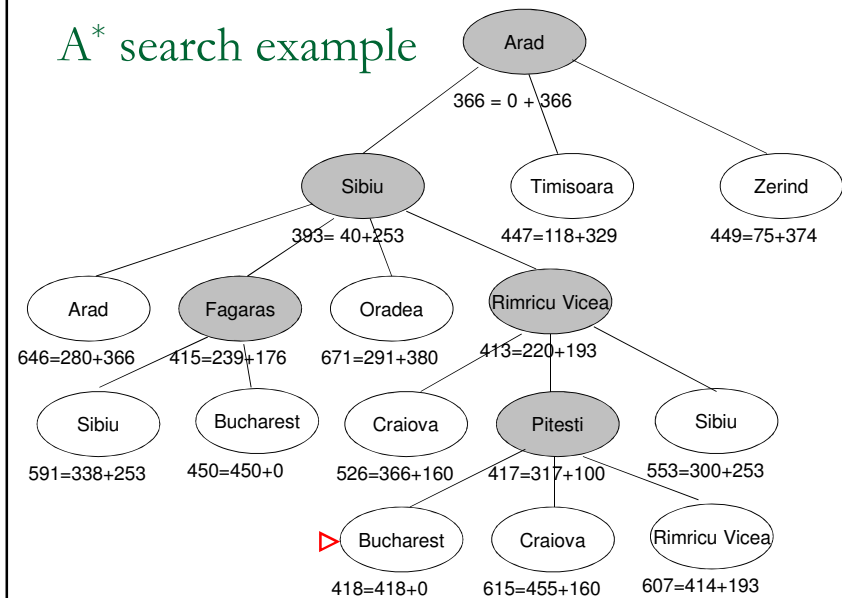
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A* search example



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A* search example



Can we Prove Anything?

- If the state space is finite and we avoid repeated states, the search is complete, but in general is not optimal
- If the state space is finite and we do not avoid repeated states, the search is in general not complete
- If the state space is infinite, the search is in general not complete

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Admissible heuristic

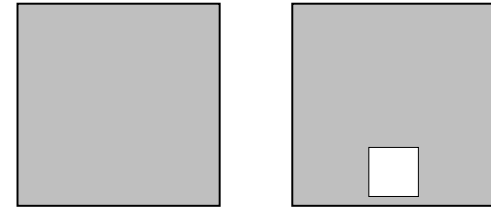
- Let $h^*(N)$ be the **true** cost of the optimal path from N to a goal node
- Heuristic $h(N)$ is **admissible** if:
$$0 \leq h(N) \leq h^*(N)$$
- An admissible heuristic is always **optimistic**

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Admissible heuristics

The 8-puzzle:

- $h_1(n)$ = number of misplaced tiles
- $h_2(n)$ = total Manhattan distance
(i.e., no. of squares from desired location of each tile)



- $h_1(S) = ?$ 7
- $h_2(S) = ?$ $2+3+3+2+4+2+0+2 = 18$

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```
ERROR: rangecheck  
OFFENDING COMMAND: get
```

```
STACK:
```

```
1  
[[-168 -341 1093 960 ]]  
0  
/descender  
[(Start State)]  
-savelevel-  
-savelevel-
```